INDONESIA E-COMMERCE SALES ANALYSIS

DATA ANALYTICS PROJECT PORTFOLIO

End-to-End Data Analytics Project

Data Preparation, Exploratory Analysis, Business Insights, and Recommendations

PROJECT PROFILE	DESCRIPTION
Project Type	Business Data Analytics
Dataset	Indonesia E-Commerce Sales
Data Source	Kaggle — zkyfauzi/indonesia-ecommerce-sales
Observation Period	1 December 2023 – 30 November 2025
Total Records	18,868 orders
Primary Tools	Python, Pandas, Excel, Google Colab

A data-driven study of order performance, payment behavior, cancellation patterns, and operational improvement opportunities.

Executive Summary

This project analyzes Indonesia e-commerce sales data to evaluate order performance, payment behavior, shipping patterns, and cancellation characteristics. The analysis uses 18,868 order records covering December 2023 to November 2025.

The dataset contains **18,868** orders, including **16,045** completed orders, **2,573** cancelled orders, and **250** orders with other statuses.

Completion rate: 85.04% | Cancellation rate: 13.64%

The analysis identifies cancellation behavior as an important operational evaluation area. Buyer-related decisions, order changes, unpaid orders, and address changes represent the four largest custom analytical cancellation themes.

The project translates these findings into seven proposed business recommendations and a KPI monitoring framework. Recommendations are analytical proposals, not experimentally proven interventions.

1. Project Objectives

• Measure overall order completion and cancellation performance.

• Evaluate payment-method performance using completion and cancellation rates.

• Examine shipping-option performance and unusual cancellation patterns.

• Identify dominant cancellation reasons and responsible actors.

• Explore relationships between order quantity, weight, payment, and shipping cost.

• Develop actionable business recommendations and KPI monitoring indicators.

2. Dataset Profile

Attribute	Description
Dataset	Indonesia E-Commerce Sales
Source	Kaggle — zkyfauzi/indonesia-ecommerce-sales
Raw records	18,868 rows
Clean dataset	18,868 rows × 22 columns
EDA dataset	18,868 rows × 23 columns
Unique order IDs	18,868
Observation period	1 December 2023 – 30 November 2025
Order date range	2023-12-01 to 2025-11-30

The clean dataset contains 22 columns. The exploratory dataset contains an additional derived order_month field for monthly analysis.

3. Data Preparation

Procedure	Treatment
Missing order dates	Reconstructed using Waktu Pesanan Dibuat.
Invalid date formats	Standardized into a consistent datetime format.
Payment method	Missing values represented as Tidak Diketahui.
Missing quantity	Retained as missing; no unsupported imputation.
Payment labels	Standardized into consistent categories.
Order identifiers	Validated for uniqueness.
Derived features	Created order_month for temporal analysis.

The cleaning process prioritizes traceability and avoids replacing unknown business values with unsupported assumptions.

4. Analytical Methodology

Stage	Method	Purpose
Data preparation	Cleaning and validation	Improve consistency and reliability.
EDA	Descriptive statistics	Understand distributions and patterns.
Performance analysis	KPI and rate comparison	Measure order and payment performance.
Cancellation analysis	Reason classification	Identify dominant cancellation patterns.
Multivariate analysis	Spearman correlation	Explore monotonic relationships.
Business analysis	Evidence-based recommendations	Translate findings into proposed actions.

5. Key Business Findings

Metric	Result
Total orders	18,868
Completed orders	16,045
Cancelled orders	2,573
Other statuses	250
Completion rate	85.04%
Cancellation rate	13.64%
Total order payment	Rp 962.091.801
Completed order payment	Rp 950.963.704
Average completed-order payment	Rp 59.268,54

Total payment represents recorded order payment values. It must not be interpreted as profit or net revenue because cost of goods sold and other cost components are unavailable.

6. Order Performance

The dataset records 18,868 orders. 16,045 orders were completed, producing a completion rate of 85.04%.

There were 2,573 cancelled orders, equivalent to 13.64% of total orders. This indicates that cancellation management is an important area for operational review.

The remaining 250 orders belong to other recorded status categories. These statuses are not necessarily failed orders.

7. Payment Performance

Payment Method	Completion Rate	Cancellation Rate
COD	84.89%	13.47%
Online Payment	80.28%	18.74%
Saldo ShopeePay	90.28%	8.85%

Online Payment has the highest cancellation rate among the three displayed payment methods. Saldo ShopeePay records the highest completion rate. These are descriptive comparisons and do not establish that payment method causes order outcomes.

8. Cancellation Analysis

The dataset contains 2,573 cancelled orders. Cancellation themes are custom analytical groupings created for this project; they are not official platform categories.

Cancellation Theme	Orders
Buyer decision	624
Order changes	603
Unpaid orders	403
Address changes	353
Shipping failure	238
Fulfillment delays	160
Out of stock	13
Other recorded themes	179
TOTAL	2,573

The four largest themes account for 1,983 cancellations or 77.07% of cancelled orders.

Cancellation Actors

Actor	Orders	Share
Buyer	1,736	67.47%
Automatic system	824	32.02%
Seller	13	0.51%

Buyer-attributed cancellations represent the largest actor group. Automatic cancellations also form a substantial portion and deserve separate operational investigation.

9. Business Recommendations

Priority	Theme	Suggested Action
High	Buyer decision	Improve product information and purchase-decision support.
High	Order changes	Review order-editing and confirmation workflows.
High	Unpaid orders	Evaluate payment reminders and payment completion friction.
High	Address changes	Improve address validation before order confirmation.
High	Shipping failure	Review failed delivery and logistics handover records.
Medium	Fulfillment delays	Monitor seller preparation and dispatch timelines.
Medium	Stock availability	Improve inventory synchronization and stock accuracy.

Priority levels are qualitative project recommendations. They are not measured financial-impact rankings and should be validated through operational experiments and subsequent monitoring.

10. Multivariate Analysis

Variable Pair	Spearman Correlation
Total quantity vs weight	0.48
Total quantity vs payment	0.54
Weight vs payment	0.75
Payment vs estimated shipping	0.58

The strongest displayed positive monotonic relationship is between weight and payment, with a Spearman correlation coefficient of 0.75.

Correlation describes association, not causation. The results do not prove that a change in one variable directly causes a change in another.

11. KPI Monitoring Framework

KPI	Measurement	Review
Completion rate	Completed orders / total orders × 100%	Monthly
Cancellation rate	Cancelled orders / total orders × 100%	Monthly
Payment completion rate	Completed orders by payment method / total orders by method	Monthly
Unpaid cancellation share	Unpaid cancellations / all cancellations × 100%	Monthly
Shipping failure share	Shipping failures / all cancellations × 100%	Monthly
Fulfillment delay share	Fulfillment delay cancellations / all cancellations × 100%	Monthly
Stock-related cancellation	Out-of-stock cancellations / all cancellations × 100%	Monthly

The KPI framework provides measurement definitions and review frequencies. Numeric targets should be established after a baseline period and operational feasibility assessment.

12. Data Limitations

• Cost of goods sold (COGS) is unavailable; profit cannot be calculated.

• Profit and ROI are not established by this dataset.

• Customer ID is unavailable; unique customer counts cannot be measured.

• Correlation does not demonstrate causation.

• Missing quantity values are retained rather than imputed without evidence.

• Products may belong to multiple categories, preventing simple order-payment allocation.

• Some November 2025 orders have other statuses, including in-transit and return-window statuses; these are not necessarily failed orders.

• December 2024 and July 2025 contain zero recorded orders; this does not prove there was no business activity.

• Return reasons are unavailable, limiting return root-cause analysis.

• Cancellation themes are project-defined groupings and may involve classification assumptions.

13. Tools & Skills Demonstrated

Area	Skills Demonstrated
Python	Data analysis and automation.
Pandas	Data transformation, aggregation, and validation.
Data cleaning	Missing-value treatment, date parsing, standardization.
Exploratory analysis	Descriptive statistics and pattern identification.
Statistical analysis	Spearman rank correlation.
Dashboard	KPI cards, charts, and analytical worksheets.
KPI framework	Metric definitions and monitoring methods.
Business analysis	Evidence-based interpretation and recommendations.

14. Project Deliverables

• Project Charter

• Data Collection and Preparation Report

• Data Cleaning Log

• Exploratory Data Analysis Report

• Root Cause Analysis Report

• Recommendation Matrix

• Final Dashboard

• Final Analysis Report

• Final Presentation

• Portfolio Project

• Clean Dataset

15. Portfolio Summary

Indonesia E-Commerce Sales Analysis is an end-to-end data analytics project covering data preparation, exploratory analysis, order and payment performance, cancellation analysis, statistical association, business recommendations, and KPI monitoring.

The project demonstrates the ability to transform raw transactional data into structured analytical outputs and communicate findings with appropriate limitations.

16. Project Integrity Statement

All reported findings are based on the analyzed dataset and the documented analytical workflow. Missing information has not been presented as known fact. Correlation is not interpreted as causation, and recorded payment values are not represented as profit or ROI.

Business recommendations are proposals derived from descriptive findings. Their actual effectiveness requires operational testing and continued KPI monitoring.

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